A Phase 1/2, Multicenter, Randomized, Placebo Controlled, Double Blind Single Dose and Multiple Dose Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of DNL593 in Healthy Participants and Participants with Frontotemporal Dementia (FTD GRN)

Study Identifier

NCT05262023

Sponsor

Denali Therapeutics Inc.

Background & Rationale

Mutations in the granulin (GRN) gene lead to reduced progranulin (PGRN), causing lysosomal dysfunction and neurodegeneration in frontotemporal dementia (FTD GRN). DNL593 is a brain penetrant PGRN replacement therapy designed to increase PGRN levels and restore lysosomal function. The Phase 1/2 study will investigate safety, tolerability, pharmacokinetics (PK), and pharmacodynamics (PD) (A Study to Evaluate the Safety, Tolerability …).

Objectives

- **Primary objective:** Evaluate safety and tolerability of single and multiple intravenous doses of DNL593 in healthy participants and participants with FTD GRN.
- **Secondary objectives:** Characterize the PK and PD of DNL593 (A Study to Evaluate the Safety, Tolerability …).
- **Exploratory objectives:** Assess biomarker changes and clinical signals of effect in participants with FTD GRN.

Study Design

This is a multicenter, randomized, placebo controlled, double blind study with three parts (A Study to Evaluate the Safety, Tolerability …):

- **Part A**: Single ascending dose in healthy participants of nonchildbearing potential (A Study to Evaluate the Safety, Tolerability …).
- **Part B**: Multiple-dose study in participants with FTD GRN over 25 weeks, evaluating safety, tolerability, PK, and PD (A Study to Evaluate the Safety, Tolerability …).
- **Part C**: Optional 18 month open-label extension (OLE) for participants who complete Part B (A Study to Evaluate the Safety, Tolerability …).

Population

Part A (Healthy Volunteers)

- Adults aged 18-55 years (A Study to Evaluate the Safety, Tolerability …).
- Women must be surgically sterilized or post menopausal; men may participate (A Study to Evaluate the Safety, Tolerability …).
- Body mass index (BMI) between 18 and 32 kg/m² (A Study to Evaluate the Safety, Tolerability …).
- Contraception requirements: when engaging in sex with a woman of child-bearing potential, two forms of birth control are required (A Study to Evaluate the Safety, Tolerability …).

Part B (Participants with FTD GRN)

- Adults aged 18-80 years (Clinical Trials Register).
- BMI between 18 and 32 kg/m² (Clinical Trials Register).
- Clinical Dementia Rating plus National Alzheimer's Coordinating Center frontotemporal lobar degeneration (CDR+NACC FTLD) global score ≥ 0.5 (Clinical Trials Register).
- Confirmed GRN mutation by genetic testing or historical record (Clinical Trials Register).
- Contraception requirement for male participants and their partners (Clinical Trials Register).

Key exclusion criteria include clinically significant neurologic, psychiatric, or medical diseases; recent stroke or seizures; head trauma with loss of consciousness within two years; and pregnancy or lactation.

Intervention

- **Investigational product:** DNL593 administered intravenously. Healthy participants receive single ascending doses; participants with FTD GRN receive multiple doses.

- **Comparator:** Placebo intravenous infusion.
- Participants are randomized to receive DNL593 or placebo in Parts A and B. The study is double-blind in Parts A and B and open-label in Part C.

Endpoints

- **Safety and tolerability:** Incidence of adverse events, serious adverse events, clinical laboratory parameters, vital signs, and ECGs.
- **Pharmacokinetics:** Plasma and CSF DNL593 levels and PK parameters (Cmax, AUC, half-life).
- **Pharmacodynamics/Biomarkers:** Changes in PGRN levels (plasma, CSF), lysosomal function markers, and neurodegeneration biomarkers.
- **Clinical exploratory:** For participants with FTD GRN, changes in cognitive and behavioral scales and functional measures over 25 weeks and in the extension.

Statistical Considerations

Descriptive statistics will summarize safety and tolerability. PK/PD modelling will be used to analyze DNL593 concentrations and biomarker responses. In Part B, comparisons between DNL593 and placebo on biomarker changes will use appropriate models (e.g., ANCOVA). Approximately 106 participants are expected across all parts.

Duration and Timeline

- Study start date: February 1 2022.
- Estimated primary completion and study completion date: November 30 2025.
- Part A: single-dose evaluation (duration defined by cohort schedule).
- Part B: 25-week multiple-dose treatment period (A Study to Evaluate the Safety, Tolerability …).
- Part C: optional 18-month open-label extension (A Study to Evaluate the Safety, Tolerability …).

Ethics and Safety Monitoring

The study will be conducted in accordance with the Declaration of Helsinki and Good Clinical Practice guidelines. Institutional Review Board or Independent Ethics Committee approvals will be obtained for each site. A Data Safety Monitoring Board (DSMB) will oversee safety data. Written informed consent is required from all participants before any study procedures. Participants and their partners must use effective contraception as specified in the inclusion criteria (A Study to Evaluate the Safety, Tolerability …, Clinical Trials Register).